Materials Science and Engineering

	Normative 4-year Degree
Major	Materials Science and Engineering 材料科學及工程
Degree / Award Title	Bachelor of Engineering in Materials Science and Engineering 工學士(材料科學及工程)
Offering Academic Unit	Department of Materials Science and Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12
College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or*	3

	MA1300 Enhanced Calculus and Linear Algebra I	
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	MSE2066 Materials Engineers in Society	3

College / School Requirements (6 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
CHEM1300	Principles of General Chemistry	3
PHY1201	General Physics I	3

Major Requirements (72 credit units) **Catalogue Term : Semester A 2025/26**

1. Core Courses (60 credit units)

Course Code	Course Title	Credit Units	Remarks
CS1302	Introduction to Computer Programming	3	
PHY1202	General Physics II	3	
MSE1001	Programme Induction	0	
MSE2102	Introduction to Materials Science and Engineering	3	
MSE2104	Mechanical Behaviour of Materials	3	
MSE2106	Quantum Properties of Materials	3	
MSE2107	Measurement Theory and Application	3	
MSE2108	Materials Chemistry	3	
MSE2109	Bonding and Structure of Materials	3	
MSE2243	Workshop Practice	3	
MSE3110	Deformation and Fracture	3	
MSE3113	Soft Materials	3	
MSE3114	Fundamentals of Scientific Computing: a Course Powered by AI	3	
MSE3171	Materials Characterization	3	
MSE3172	Electronic Properties of Solids	3	
MSE3190	Thermodynamics of Materials	3	
MSE3195	Kinetics and Phase Transformations	3	
MSE3244	Design Laboratory	3	

Select **one** from the following:

Course Code	Course Title	Credit Units	Remarks
MSE4116	Dissertation	6	
FS4003	CES Placement Project	6	

Select **one** from the following:

Course Code	Course Title	Credit Units
MA2001	Multi-variable Calculus and Linear Algebra	3
MA2158	Linear Algebra and Calculus	3
MA2181	Mathematical Methods for Engineering	3

2. Electives (12 credit units)

Course Code	Course Title	Credit Units
MA2172	Applied Statistics for Sciences and Engineering	3
MA2177	Engineering Mathematics and Statistics	3
MSE3111	Ceramic Materials	3
MSE3130	Biomaterials	3
MSE4121	Thin Films	3
MSE4170	Corrosion and Protection of Materials	3
MSE4118	Composites	3
MSE4119	Modeling Materials: a Course Powered by AI	3
MSE4124	Failure Analysis and Case Studies	3
MSE4178	Nanostructures and Nanotechnology	3
MSE4179	Advanced Materials Characterization and its Industrial Applications	3
MSE4307	Building Materials	3
MSE4714	Special Topics in Materials Science and Engineering I	3
MSE4715	Special Topics in Materials Science and Engineering II	3
FS4002	Industrial Attachment Scheme	3
FS4005	Overseas Internship Scheme	3